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REVIEW ARTICLE

Dermatologic Toxicities Induced by JAK Inhibitors

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Abstract

Janus kinase inhibitors are increasingly used for the treatment of a wide range of dermatologic and non-dermatologic immune-mediated diseases, leading to growing interest in their safety profile, particularly with respect to cutaneous adverse events. This narrative review summarizes current evidence on dermatologic toxicities associated with both systemic and topical Janus kinase inhibitors. The most frequently reported cutaneous adverse event is an acne-like eruption, which shows a clear dose-dependent pattern and typically occurs early after treatment initiation. Cutaneous infections represent another major group of adverse events, with herpes zoster being the most clinically relevant. Less frequently, cutaneous malignancies have been reported, predominantly non-melanoma skin cancers, with a stronger signal observed in hematologic populations and in patients treated with ruxolitinib. Overall, dermatologic adverse events associated with Janus kinase inhibitors are usually manageable and rarely require permanent treatment discontinuation. Increased awareness, early recognition, and appropriate dermatologic management are essential to minimize morbidity and to support long-term treatment adherence across clinical settings.

1 Introduction

Janus kinase (JAK) inhibitors are a class of small-molecule targeted drugs that interfere with intracellular cytokine signaling through inhibition of the JAK signal transducer and activator of transcription (STAT) pathway, which plays a key role in immune-mediated and inflammatory diseases [1–3]. The JAK family includes four kinases, JAK1, JAK2, JAK3, and tyrosine kinase 2 (TYK2), which associate with multiple cytokine receptors and mediate downstream transcription of genes involved in inflammation, immune activation, and cell survival [1, 2]. Because more than 50 cytokines signal through only four JAK proteins, JAK inhibition allows simultaneous modulation of several inflammatory pathways, explaining both the broad efficacy and the pleiotropic safety profile of these agents,

Key Points

Janus kinase inhibitors are associated with a characteristic spectrum of dermatologic adverse events across multiple medical specialties (dermatology, gastroenterology, rheumatology, hematology).

Acne-like eruption (called JAKne) is the most common cutaneous toxicity and is typically dose dependent and early in onset. Cutaneous infections, particularly herpes zoster, represent a relevant class effect of systemic Janus kinase inhibitors. Non-melanoma skin cancers are uncommon in dermatologic patients treated with Janus kinase inhibitors but may occur more frequently in hematologic populations receiving these therapies.

Most dermatologic adverse events are manageable and rarely necessitate permanent treatment discontinuation.

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as consistently described in multiple narrative reviews and mechanistic studies [1, 2].

Over the last decade, JAK inhibitors have been approved or extensively used for a wide range of dermatologic and non-dermatologic conditions, including rheumatoid arthritis (RA), psoriatic arthritis, ankylosing spondylitis, inflammatory bowel disease (IBD), alopecia areata (AA), atopic dermatitis (AD), vitiligo, myeloproliferative disorders, SAPHO syndrome, and severe acute coronavirus disease 2019 infection, as documented by multiple randomized clinical trials, long-term extension studies, and systematic reviews [4–17]. This progressive expansion across medical specialties has resulted in increasing exposure of heterogeneous patient populations to JAK inhibitors, raising clinically relevant concerns regarding adverse events (AEs) across different organ systems, including the skin, as highlighted by regulatory warnings and post-marketing safety analyses [18].

In dermatology, JAK inhibitors have profoundly changed the therapeutic options of inflammatory cutaneous diseases, particularly AA and AD, where rapid onset of action and high efficacy have been consistently demonstrated in phase III trials, real-world cohorts, and integrated safety analyses [13, 19, 20]. Baricitinib, a selective JAK1/2 inhibitor, has been approved for severe AA at a dose of 2 or 4 mg once daily, based on pivotal randomized trials and supported by real-world evidence confirming sustained efficacy and an overall manageable safety profile [13, 20–22]. Ritlecitinib, a selective JAK3/TEC family inhibitor, has also demonstrated efficacy in AA, with integrated analyses confirming its clinical effectiveness and safety in both randomized trials and real-world settings [12]. Other JAK inhibitors, including tofacitinib and ruxolitinib, have been widely used in AA and other inflammatory dermatoses, contributing substantially to the characterization of cutaneous AEs in both clinical trials and observational studies [23, 24].

In AD, oral JAK inhibitors such as abrocitinib, baricitinib, and upadacitinib have shown significant improvements in disease severity and pruritus at doses that range from 100 mg to 200 mg for abrocitinib, 2–4 mg for baricitinib, and 15–30 mg for upadacitinib [14, 15, 19, 25–27]. Across pivotal trials and real-world cohorts, dermatologic AEs such as acne, folliculitis, herpes simplex virus (HSV) infection, and herpes zoster have emerged as the most frequently reported cutaneous AEs (Table 1), often showing a dose-dependent relationship, and represent one of the most common reasons for treatment interruption in routine clinical practice [8, 15, 24–43].

Beyond approved dermatologic indications, JAK inhibitors have been investigated in other inflammatory dermatoses, including lichen sclerosus, lichen simplex chronicus, frontal fibrosing alopecia, granulomatous dermatoses, immune-mediated photodermatoses, and inflammatory nail disorders [44–51]. In these settings, dermatologic AEs were generally mild and manageable.

Topical JAK inhibitors, particularly ruxolitinib cream, have shown a favourable cutaneous safety profile, with local irritation, erythema, and folliculitis being the most frequently reported events across clinical trials and off-label experiences [17, 52, 53].

Importantly, dermatologic AEs of oral JAK inhibitors have also been widely reported in non-dermatologic indications, including RA, psoriatic arthritis, ankylosing spondylitis, IBD, hematologic malignancies, and acute coronavirus disease 2019 infection [4, 6, 7, 9]. A broad spectrum of cutaneous infections has been reported, including viral infections (e.g., herpes zoster and herpes simplex), bacterial infections (e.g., cellulitis, impetigo and folliculitis), and fungal infections (e.g., dermatophytosis

Table 1 Dermatological safety data from the pivotal phase III trials conducted with Janus kinase inhibitors used in dermatology

Manifestations	Upadacitinib [8, 15, 29–31]	Baricitinib [32–37]	Abrocitinib [26, 38–40]	Deucravacitinib [41–43]
Acne	7–17%[a]	4.7–5.8%	1.6–6.6%[b]	4–5.6%
Herpes zoster	0.9–7%	0.7–7%	0.8–3.4%	0.7–1.2%
Herpes simplex	0.9–7%	1–4%	0.9–1.3%[c]	NP
Oral herpes	3–8%	1.4%	1–2%[4]	2%
Eczema herpeticum	1%	NP	1%	NP
Non-melanoma skin cancer	< 1%	0	< 1%	< 1%

[a]Higher rate of acne was observed with upadacitinib 30 mg vs 15 mg

[b]Higher rate of acne was observed with abrocitinib 200 mg vs 100 mg

[c]Skin, oral and ophthalmic herpes simplex virus infections (together): 6%

[d]Genital herpes simplex virus superinfection: 0.7%

NP not provided

and candidiasis) [8–10, 54]. Neoplastic skin conditions, including non-melanoma skin cancer and Kaposi sarcoma, have been reported, particularly in patients with hematologic diseases or additional immunosuppressive risk factors [1, 18, 55–57]. These observations likely reflect differences in baseline immune status, cumulative immunosuppression, dosage, and treatment duration across clinical settings, as highlighted in safety-focused dermatologic reviews [1, 58–61].

More recently, JAK inhibitors have also been evaluated in connective tissue diseases such as dermatomyositis, a condition characterized by an increased baseline risk of malignancy. In a recent phase III randomized trial, brepocitinib, a selective TYK2/JAK1 inhibitor, demonstrated significant efficacy. Serious infections were more frequent with the 30-mg dose (10%) compared with placebo (1%), highlighting a dose-dependent safety signal. Given the intrinsic malignancy risk associated with dermatomyositis and the immunomodulatory effects of JAK inhibition, long-term safety, particularly with respect to malignancy and infection risk, warrants careful monitoring in these populations [62].

Given the widespread and increasing use of JAK inhibitors across dermatology, rheumatology, gastroenterology, and hematology, dermatologists play a central role in the early recognition, diagnosis and management of JAK inhibitor-associated AEs [1, 2]. For these reasons, we conducted a narrative review of the literature, focusing on dermatologic toxicities associated with JAK inhibitors used in both dermatologic and non-dermatologic conditions. The aim of this review is to summarize current evidence on the type, frequency, and clinical relevance of cutaneous AEs reported with JAK inhibitors, highlighting similarities and differences across indications and providing practical insights for clinicians involved in multidisciplinary patient care.

2 Materials and Methods

A structured narrative literature review was conducted by five authors using MEDLINE/PubMed as the primary database, covering publications until March 2026. The review was designed to address the spectrum, incidence, and clinical relevance of dermatologic AEs associated with systemic and topical JAK inhibitors across different medical settings. The search strategy combined Medical Subject Headings (MeSH) terms and free-text keywords related to JAK inhibition and cutaneous toxicity, including combinations of: "Janus kinase inhibitors," "JAK inhibitors," "baricitinib," "upadacitinib," "abrocitinib," "tofacitinib," "ruxolitinib," "ritlecitinib," "deucravacitinib," "dermatologic adverse events," "cutaneous adverse events," "skin toxicity," "acne," "acneiform rash," "JAKne," "herpes simplex," "herpes zoster," "eczema herpeticum," "molluscum contagiosum," "epidermodysplasia verruciformis," "*Staphylococcus aureus*," "cellulitis," "impetigo," "bacterial infection," "cutaneous infection," dermatophytosis," "fungal infection," "candidiasis," "mycobacteria," "skin cancers," "squamous cell carcinoma," "basal cell carcinoma," "Kaposi sarcoma," and "malignancies". The final selection of articles reflects a comprehensive and multidisciplinary synthesis of the available literature, aimed at providing a clinically oriented overview of dermatologic toxicities associated with JAK inhibitors in both dermatologic and systemic disease settings. This review is based on previously published articles and does not contain any new studies with human participants or animals performed by any of the authors.

3 Acne-Like Eruption (or JAKne)

Among dermatologic AEs observed during treatment JAK inhibitors, acne or acne-like eruption, often referred to "JAKne," represent the most frequently reported cutaneous toxicity across dermatologic and non-dermatologic indications [63–66]. Randomized controlled trials, real-world observational studies, and systematic reviews consistently identify acne as the most common treatment-emergent dermatologic AE associated with systemic JAK inhibitors (Table 2) [1, 25, 64, 66].

3.1 Incidence and Risk Across Janus Kinase (JAK) Inhibitors

The overall incidence of JAKne is estimated at about 5–10%, with a relative risk of 3.8 [67, 68]; however, the incidence varies significantly according to the specific JAK inhibitor, dosage, and underlying disease [69]. In a recent network metanalysis the incidence was ranked as follows: JAK1 inhibitors > TYK2 inhibitors > combined JAK1 and JAK2 inhibitors > combined JAK1 and TYK2 inhibitors > JAK3 + TEC inhibitors > pan-JAK inhibitors [69]. Meta-analyses of randomized controlled trials in AD demonstrated a markedly increased risk of acne with abrocitinib and upadacitinib, whereas baricitinib is associated with a substantially lower risk [61, 70]. In pooled analyses, the relative risk of acne reaches approximately five-fold with abrocitinib and upadacitinib compared with placebo, while no statistically significant increase is observed with low-dose baricitinib [25]. Dose dependency represents a key feature of JAKne, with higher rates consistently reported at higher daily doses of upadacitinib (30 mg vs 15 mg) and abrocitinib (200 mg vs 100 mg) [25, 65, 70–73].

Table 2 JAKne: main clinical data [25, 63, 67, 68, 72–74]

JAKne: main clinical data	
Epidemiological data	Overall incidence: 5–10% Pooled odds ratio and relative risk: 3.8
	Higher incidence with upadacitinib (15%) and possibly abrocitinib (dose-dependent mechanism)
	More frequent in women and in patients aged 20–50 years
	New-onset acne in >90% of cases
	• Other risk factors: • Personal or family history of acne • Selective TYK2, JAK1 or JAK 1/2 inhibitors • Dermatological conditions treated (e.g., atopic dermatitis)[a]
Clinical presentation	Time to onset (mean): 6 weeks[b]
	Mild-to-moderate severity
	Face (>90%), chest/back (20–30%), arms (<5%)
	Mainly inflammatory papules and pustules (>90%), open and closed comedones (40%), nodules and cysts are uncommon
	Pruritus may occur
	Variable negative impact on quality of life (psychological distress)
Management	Dermatological management required in 30–60% of exposed patients (especially skin-directed therapies)
	Changes in JAK inhibitor therapy (dose reduction, interruption, discontinuation) in 18–25%
	Scarring or post inflammatory pigmentation in <10%
	Time to resolution (mean): 85 days
	Low risk of recurrence

JAK Janus kinase, *TYK* tyrosine kinase

[a]Possibly as a consequence of increased awareness and diagnostic sensitivity among dermatologists

[b]Late-onset JAKne has been also described (after 6 months or more)

3.2 Time of Onset

JAKne typically develops early after treatment initiation, representing a characteristic and clinically useful feature of this AE. Across different disease settings, it most commonly appears within the first 6 weeks of therapy (70% of cases), usually during the induction phase rather than during long-term maintenance [71, 74]. In real-world cohorts of Italian patients with moderate-to-severe AD treated with upadacitinib, the vast majority of cases occurring within the first 6 months of treatment [63]. Similar findings have been reported in Japanese populations, where acne was among the earliest treatment-emergent AE, often developing within the first months of therapy [71]. In adolescent patients, clinical trials have shown a comparable temporal pattern, with acne usually emerging within the first 8 weeks of exposure and rarely reported after the initial treatment period [14]. However, the occurrence of late-onset acne (after 6 months or more) has been also sporadically reported [63].

Overall, the consistent early onset across different populations supports a direct pharmacologic effect of JAK inhibition and highlights the importance of close dermatologic monitoring during the first weeks or months of therapy [74].

3.3 Patient-Related Risk Factors

Several patient-related factors influence the risk of developing JAKne. Younger age represents a major risk factor, with adolescents and young adults showing the highest incidence, although new-onset acne has also been observed in older patients without previous acne history [14, 71, 74]. Female sex has been consistently associated with an increased risk of JAKne across both dermatologic and gastroenterologic cohorts [63, 67, 71, 73–75]. Ethnicity also appears to modulate clinical expression, with non-white patients experiencing a higher incidence and a greater risk of post-inflammatory hyperpigmentation [75]. Concomitant use of topical or systemic corticosteroids further increases susceptibility to JAKne, particularly in patients with AD and IBD [71, 74, 75].

Notably, most patients developing JAKne have no personal history of acne vulgaris, supporting the concept of a distinct drug-induced entity rather than an exacerbation of pre-existing acne [63, 75]. However, a personal or familial history of acne appears to be an independent risk factor for the development of more severe forms of JAKne [72, 73].

3.4 Clinical Presentation

JAKne typically presents as a monomorphic or mildly polymorphic eruption characterized by inflammatory papules and pustules, with fewer comedones compared with acne vulgaris [61, 72, 76, 77]. Lesions predominantly involve the face (> 90%), particularly the cheeks, forehead, and perioral area, but may also affect the chest, upper back, and less commonly, the scalp (Fig. 1) [64, 72, 75]. Nodules and cysts are uncommon but have been reported in moderate-to-severe cases, especially at higher drug doses [73]. Most cases are classified as mild to moderate in severity, while severe forms requiring systemic therapy remain relatively rare [20, 68, 72, 73].

Itching could be an associated symptom in inflammatory lesions, particularly in patients with underlying AD, where itch represents a baseline disease feature that may be exacerbated by follicular inflammation [63, 72]. Several studies report that patients describe acne lesions as pruritic rather than painful, which may further contribute to excoriations and secondary infection: *Staphylococcus aureus* has been specifically reported, particularly in patients with AD and impaired skin barrier function [1, 63, 75]. Mean time to resolution was estimated at 80–100 days, with a low incidence of recurrence [72, 73].

3.5 Differential Diagnosis

Differential diagnosis with acne vulgaris is essential, as JAKne often lacks a significant comedonal component, has an abrupt onset temporally related to drug initiation, and occurs in patients outside the typical age range or without previous acne history [64, 76]. Other differential diagnoses include steroid-induced acne, rosacea-like dermatitis, and folliculitis with bacterial superinfection, which may coexist or overlap in immunomodulated patients [64].

3.6 Pathogenesis

The pathogenesis of JAKne remains incompletely understood but appears to be multifactorial and largely independent of classic androgen-mediated mechanisms [25, 76]. Inhibition of the JAK-STAT pathway alters keratinocyte proliferation and follicular keratinization, potentially leading to follicular obstruction and inflammation [70, 75]. Alterations in skin microbiome composition, including changes in *Cutibacterium acnes* and *Demodex* colonization, have been hypothesized as contributory factors, particularly in patients with AD [25, 70, 75].

The strong dose dependency supports a causal relationship between JAK inhibition and acne development [61, 70].

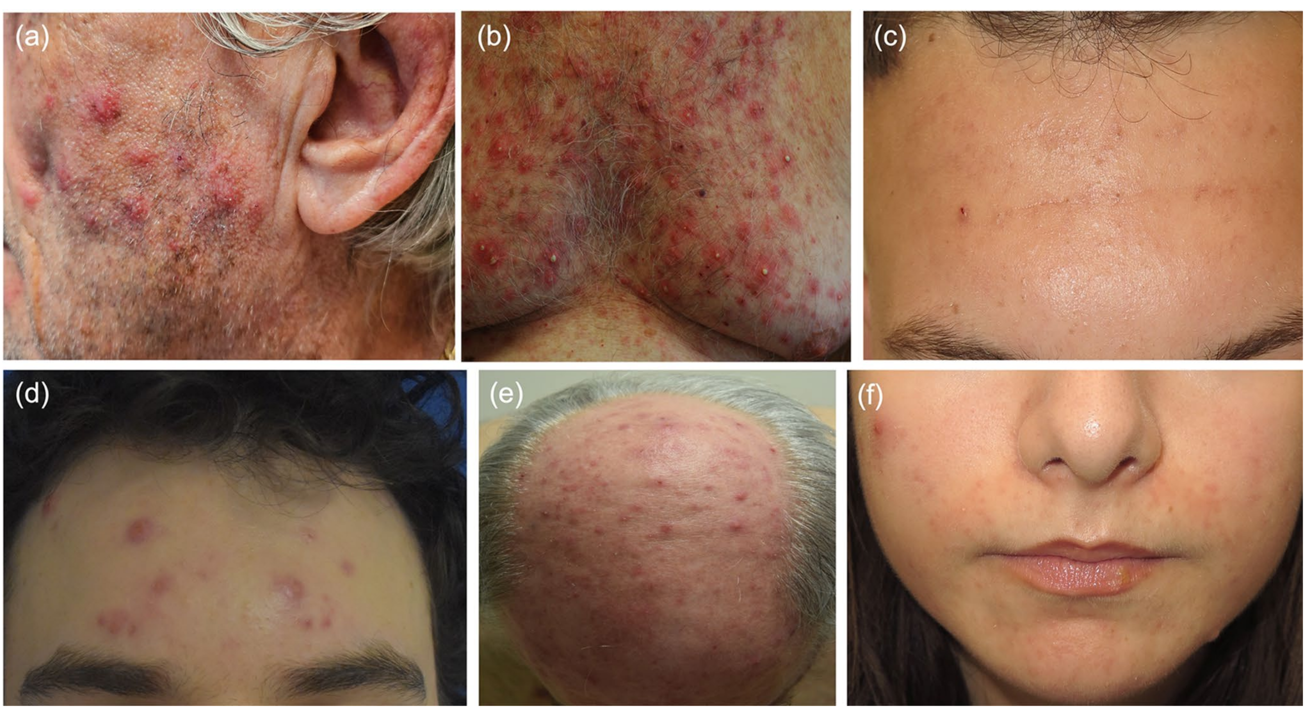

Fig. 1 Appearance of acneiform rash on the face (**a**) and on the trunk (**b**) in a 78-year-old patient undergoing oral ruxolitinib therapy for polycythaemia vera. Typical presentation of acne during baricitinib for atopic dermatitis (**c**). Acne flare-up in a 22-year-old patient 1 month after the introduction of upadacitinib 30 mg daily for atopic dermatitis, successfully treated with oral doxycycline (**d**). Papular inflammatory form of the scalp (**e**). Acneiform eruption of the face in a patient who received abrocitinib for atopic dermatitis (**f**). Personal files

Causality is also supported by improvement or resolution following dose reduction or drug discontinuation, and in several cases by recurrence upon rechallenge [72, 76].

3.7 Management

Management of JAKne should be graded by severity, and aimed at maintaining adherence to an otherwise highly effective systemic therapy because most cases can be controlled without discontinuing the JAK inhibitors [65]. A practical first step is pre-treatment counseling, informing patients that acneiform eruptions are common, usually mild-to-moderate, and typically manageable with standard acne treatments, which reduces anxiety and improves persistence on therapy [65, 72]. Early recognition during the first months of therapy is important because prompt treatment may prevent worsening, post-inflammatory sequelae, and unnecessary dose changes [63, 65].

For a mild form of JAKne, a stepwise topical approach starting with non-comedogenic skincare, gentle cleansers, and moisturizers together with a close clinical monitoring may be sufficient. However, association with anti-inflammatory topical agents may be required in about 40–-60% of cases, depending on the patient's condition [65, 72, 73]. Topical benzoyl peroxide (as monotherapy or combined), topical antibiotics (e.g. clindamycin), and azelaic acid are commonly used first-line options, particularly when lesions are mainly papulopustular [65, 76, 78]. In patients with AD, topical regimens should be chosen with attention to tolerability and barrier impairment, and irritation should be actively monitored during initiation and escalation [63, 75]. Topical retinoids may be effective in selected patients, but in eczematous skin they may aggravate dryness and irritation, therefore they often require cautious introduction (low frequency, short-contact, or alternative agents) [63, 75].

For moderate-to-severe disease, escalation to systemic therapy is sometimes needed, and oral tetracyclines (e.g., doxycycline, lymecycline, or minocycline) are commonly used because of anti-inflammatory activity and good tolerability [63, 65, 72, 73, 78]. Real-world cohorts show that most patients improve with topical therapy alone or with the addition of oral antibiotics [63, 72]. Oral isotretinoin is rarely required but may be considered in refractory cases, although a careful risk-benefit assessment is mandatory, particularly in patients with xerosis and eczema [79].

When secondary infection is suspected, particularly in patients with AD with crusting, oozing, or extensive pustules, targeted anti-staphylococcal therapy (topical or systemic) may be required, and microbiological sampling/swab can be considered in refractory cases [75]. Acne can be managed in most cases without dose reduction or treatment interruption of JAK inhibitors, and that dose modification should be reserved for severe, refractory, or highly distressing cases after adequate acne-directed therapy [63, 65, 72]. However, dose changes for management of JAKne were reported in both post-hoc integrated analysis and large retrospective series, with a 20% global incidence [72, 73].

In gastroenterology practice, a practical algorithm is to treat mild-to-moderate JAKne promptly and involve dermatology when lesions are extensive, refractory, scarring, or strongly impacting quality of life, to avoid premature discontinuation of an effective IBD drug [72].

3.8 Impact on Quality of Life

Although JAKne is usually mild to moderate in severity, it may significantly impact quality of life, self-esteem, and treatment adherence, particularly in young patients and women [64, 72, 73]. Post-inflammatory hyperpigmentation and, rarely, scarring have been reported, especially in patients with darker skin phototypes [72, 73].

4 Infections

Janus kinase inhibitors increase susceptibility to infections by impairing cytokine-mediated immune responses essential for antiviral, antibacterial, and antifungal host defense, including interferon, interleukin, and growth factor signaling pathways (Table 3) [1, 80, 81]. Across randomized trials, real-world studies and meta-analyses, infections represent one of the most frequent AEs associated with JAK inhibitor therapy, with a spectrum that ranged from mild self-limited infections to serious and opportunistic infections [20, 82, 83].

Table 3 Main risk factors for infections during JAK inhibitor therapy

Domain	Key risk factors
Patient related	Older age; comorbidities (e.g., diabetes mellitus, chronic lung disease); prior history of recurrent infections
Disease related	Underlying immune-mediated disease; high disease activity; impaired skin barrier (especially AD)
Treatment related	Higher JAK inhibitor dose; concomitant corticosteroids or immunosuppressants; prior biologic therapies
Immunologic factors	Lymphopenia; impaired interferon signaling and cell-mediated immunity
Specific considerations	History of herpes zoster or HSV; lack of vaccination (e.g., VZV)

AD atopic dermatitis, *HSV* herpes simplex virus, *JAK* Janus kinase, *VZV* varicella zoster virus

Serious infections, defined as infections requiring hospitalization, parenteral antimicrobial therapy, or leading to significant morbidity, occur more frequently in patients treated with JAK inhibitors compared with placebo and, in some settings, compared with biologic agents, particularly in RA and IBD populations or for specific infections [28, 81]. In dermatologic indications such as AD and AA, the absolute risk of serious infections is lower, but vigilance is required, especially in older patients, those with comorbidities, or those receiving concomitant systemic corticosteroids or other immunosuppressive drugs [80, 83].

4.1 General Preventive and Monitoring Recommendations

Before initiating JAK inhibitor therapy, patients should perform a clinical evaluation for previous serious or recurrent infections and be screened according to local recommendations for tuberculosis, hepatitis B and C, and other relevant infections [84]. Patients should be educated to promptly report signs and symptoms of infection, particularly vesicular eruptions, fever, and localized pain [1]. During treatment, mild infections can usually be managed without treatment suspension, moderate-to-severe infections require temporary or permanent discontinuation depending on clinical severity, patient physical status, and microbiological results [28].

4.2 Herpes Zoster

Herpes zoster represents the most clinically relevant infectious complication associated with JAK inhibitor therapy and should be considered a class effect (Table 1) [1, 57, 61, 80, 85, 86]. The relative risk was recently evaluated between 1.77 and 2.23 in atopic patients [25]. There is a significantly increased incidence of herpes zoster in patients receiving JAK inhibitors compared with placebo or biologic therapies (e.g., anti-tumor necrosis factor [TNF], interleukin-4/interleukin-13/interleukin-17 inhibitors) [81, 82, 87]. However, there is no increase in the risk of herpes zoster according to the type of JAK inhibitor. Moreover, reactivation of herpes zoster with JAK inhibitors does not appear related to a dose-dependent mechanism but can appear at any time [25].

Pathogenetically, JAK-dependent immune functions are implicated in different steps of the life cycle of varicella zoster. Direct inhibition of JAK pathway impairs type I and II interferon signaling, which is crucial for maintaining varicella-zoster virus latency, thereby facilitating viral reactivation [88]. Risk factors include older age, history of herpes zoster, Asian ethnicity, lymphopenia, and concomitant systemic corticosteroid use [82, 85, 88].

Clinically, treatment emergent herpes zoster usually presents as a vesicular eruption involving a single dermatome, most frequently located thoracic or cranial dermatomes, while multi-dermatomeric or disseminated forms with systemic involvement are uncommon but have been reported (Fig. 2) [1, 89–92]. In localized, uncomplicated herpes zoster, temporary interruption of the JAK inhibitor is recommended until complete crusting of lesions and resolution of systemic symptoms, after which therapy can usually be safely resumed. In disseminated, ophthalmic, or severe herpes zoster, JAK inhibitor therapy should be immediately discontinued and systemic antiviral therapy initiated, with reintroduction considered only after full recovery and a careful risk-benefit reassessment [25, 28, 84].

Vaccination with the recombinant zoster vaccine (Shingrix®) is strongly recommended prior to the initiation of systemic JAK inhibitor therapy, according to local treatment guidelines [56, 81, 93]. If the recombinant zoster vaccine is not available, the live attenuated zoster vaccine should be administered at least 3–4 weeks before therapy initiation [61, 94]. In selected high-risk patients (e.g., history of recurrent herpes zoster or multiple risk factors), antiviral prophylaxis may also be considered, although evidence remains limited [95].

4.3 Herpes Simplex

Herpes simplex virus infections, including oral and genital reactivations (Fig. 3), are reported with increased frequency during JAK inhibitor therapy versus placebo (Table 1), particularly in patients with AD [24, 26, 56, 80, 96]. Lesions remain mild to moderate in most cases. In addition, most patients experience a single event. Oral/perioral herpes simplex reactivation represents a frequent location in this context [38, 81, 90, 97]. Of note, a dose-dependent mechanism has been recently suggested with several JAK inhibitors [88, 97].

Mild HSV reactivations can generally be treated with standard antiviral therapy without discontinuation of the JAK inhibitor. In cases of severe HSV infection, JAK inhibitor therapy should be temporarily suspended and resumed only after complete clinical resolution [24, 83, 90, 98]. Prophylaxis should be considered for patient with frequent flares [56].

Eczema herpeticum represents a potentially severe manifestation requiring prompt recognition in atopic patients [98]. However, only a minority of patients exposed to JAK inhibitors develops eczema herpeticum, most often patients with a poor disease control. In general, a low percentage of body surface area is involved. Systemic antiviral agents are usually required [90, 97].

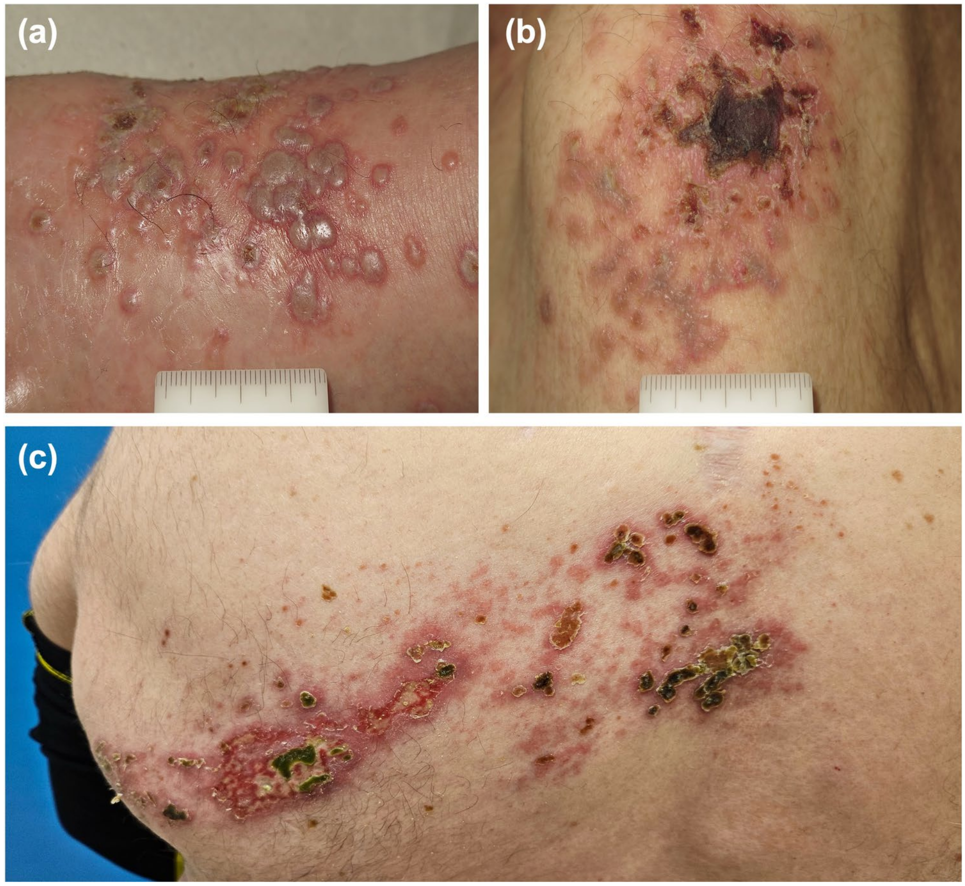

Fig. 2 Herpes zoster on the right lower limb in a patient treated with ruxolitinib (**a**). Necrotic herpes zoster in patient treated with baricitinib (**b**). Thoracic dermatomeric herpes zoster in a patient treated with ruxolitinib (**c**). Personal files

Fig. 3 Recurrent herpes simplex type 1 in a patient treated with ruxolitinib (**a**, **b**, **c**). Herpes simplex type 1 in a patient treated with ritlecitinib (**d**). Personal files

4.4 Other Viral Infections

Disseminated or localized molluscum contagiosum has been sporadically described in association with baricitinib, upadacitinib, or ruxolitinib (Fig. 4), with a complete regression after drug interruption [99–101]. Occurrence of acquired epidermodysplasia verruciformis is exceptional with JAK inhibitors (Fig. 5).

4.5 Bacterial Infections

Bacterial infections associated with JAK inhibitors are usually mild and include upper respiratory tract infections, urinary tract infections, and skin and soft-tissue infections [1, 20, 80, 97]. In most cases, in dermatologic patients, bacterial skin infections are often related to *S. aureus* colonization and barrier dysfunction rather than profound systemic

Fig. 4 Plane warts in a patient treated with ruxolitinib (**a**) and molluscum contagiosum of the pubic region in a patient treated with tofacitinib (**b**). Personal files

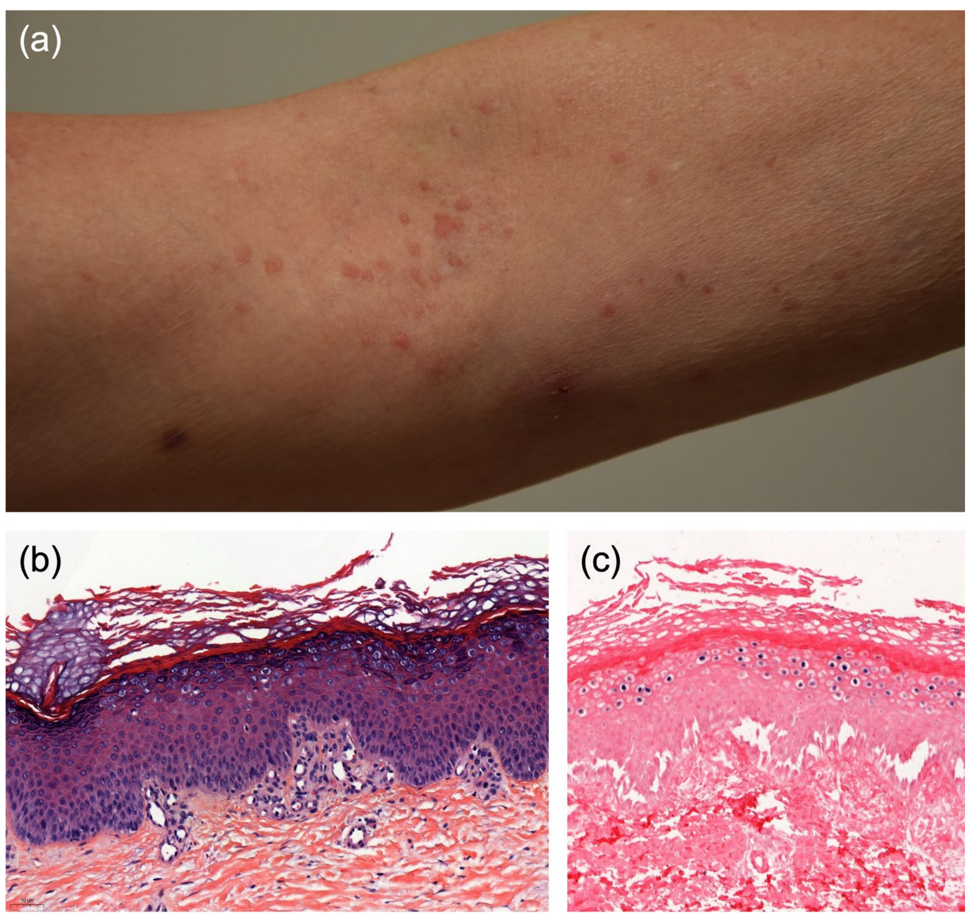

Fig. 5 Patient with epidermodysplasia verruciformis unmasked by long-term ruxolitinib therapy. Clinical image showing verrucous papules in the antecubital fold (**a**). Histology reveals a hyperkeratotic, parakeratotic, moderately acanthotic epidermis with thickened and irregular granular layer [hematoxylin and eosin stain, 20 ×] (**b**). In situ hybridization is positive for high-risk human papillomavirus (HPV) types [Ventana probes: HPV 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, and 66; 20 ×] (**c**). Personal files

immunosuppression (Fig. 6). Bacterial infections such as furunculosis, folliculitis, or impetigo can be observed, which are usually limited. However, more serious skin reactions can sometimes be noted, including erysipelas or cellulitis [80, 97].

Mild bacterial infections generally do not require JAK inhibitor interruption and can be managed with standard antimicrobial therapy. Severe or systemic bacterial infections warrant temporary discontinuation of JAK inhibitor therapy until infection resolution [20, 24, 28]. Finally, the development of cutaneous non-tuberculous mycobacteria have been rarely reported in patients with immunosuppression and exposed to tofacitinib [102, 103].

4.6 Fungal Infections

Fungal infections during JAK inhibitor therapy are uncommon and are usually superficial, including candidiasis and dermatophytosis (Fig. 7) [21, 80, 82]. Invasive fungal infections are rare and primarily reported in patients with additional immunosuppressive risk factors [28, 84].

5 Cutaneous Malignancies

The potential association between JAK inhibitors and malignancy has attracted increasing attention in recent years, largely driven by signals emerging from specific high-risk

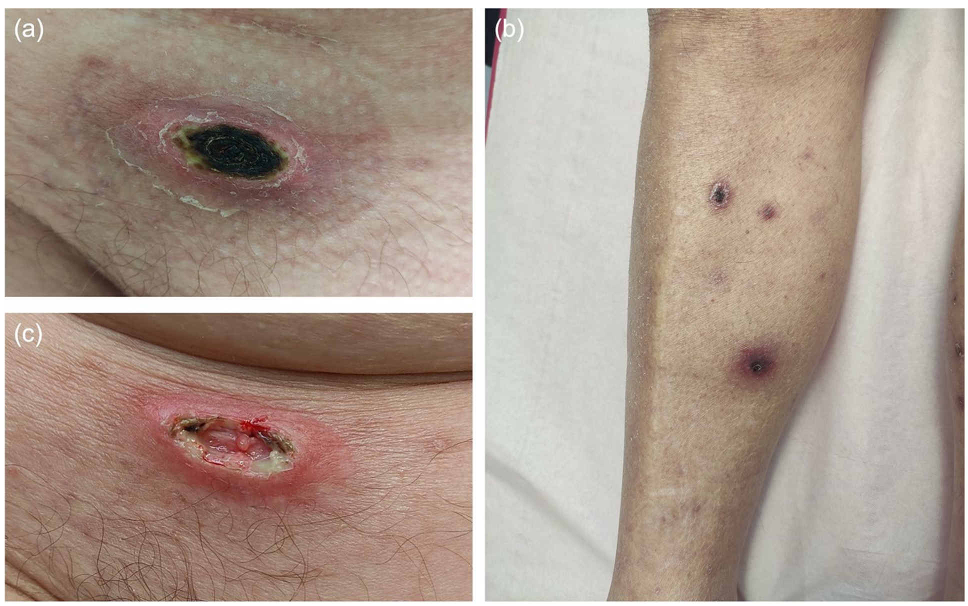

Fig. 6 Cutaneous bacterial infection of the abdomen and right leg caused by *Staphylococcus aureus* induced by baricitinib therapy for rheumatoid arthritis, before (**a**, **b**) and after specific antibiotic therapy (**c**). Personal files

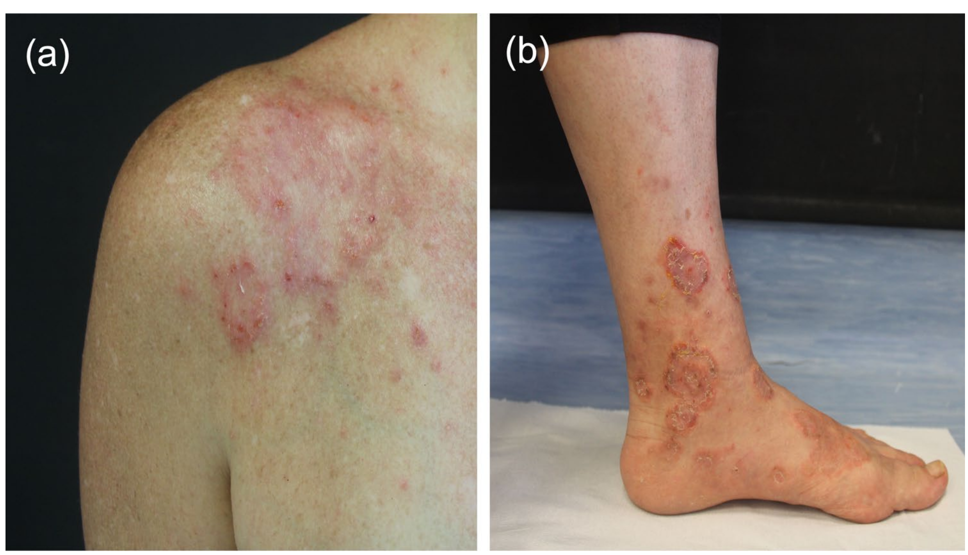

Fig. 7 Dermatophytosis of the shoulder region in a patient with atopic dermatitis receiving upadacitinib (**a**) and dermatophytosis of the right leg and foot in a patient receiving baricitinib (**b**). Courtesy of Dr. Pietro Sollena, Fondazione Policlinico Universitario Agostino Gemelli IRCCS, Rome, Italy

populations and long-term safety analyses. Preclinical evidence concerning the potential role of the JAK/STAT pathway in oncogenesis remain unclear and conflicting. The JAK/STAT signaling pathway plays a critical role in cellular growth and survival. Dysregulation or constitutive activation of this pathway may promote tumor growth, immune evasion, and apoptosis disruption. Persistent inhibition of the JAK/STAT pathway may impair natural killer cell function, in particular elimination of cancer cells [104, 105]. For example, constitutive activation of JAK2-STAT3 signaling has been documented in myeloproliferative neoplasms, and in some solid tumors, However, a large range of JAK-related cytokines has been involved in both tumor-promoting and tumor-suppressive immune responses. Moreover, it has been recently suggested that JAK1/2-STAT1/3 inhibition may display strong synergistic antitumoral activity with immune checkpoint inhibitors, potentially overcoming immunotherapy treatment resistance [106, 107].

Finally, the cancer risk between different JAK inhibitors may differ depending on their selectivity. For example, TYK2 inhibitors block interferon-gamma but have limited effects on interferon-alpha signaling and may have different effects on tumor response compared with other JAK inhibitors [108].

Cutaneous malignancies have been repeatedly discussed as a relevant safety outcome, particularly in the context of non-melanoma skin cancer (NMSC), while evidence regarding melanoma remains more limited and heterogeneous [1, 10, 59]. Across immune-mediated inflammatory diseases, the overall incidence of malignancy in patients treated with JAK inhibitors is relatively low, although interpretation is complicated by differences in baseline cancer risk, comparator therapies, duration of exposure, associated immunosuppressive therapies, and underlying disease-related immune dysregulation [10, 60]. In this context, skin cancers emerge as a distinct subgroup of interest, partly because of their known association with chronic immunosuppression and impaired immune surveillance [58, 109, 110].

In addition, inhibition of JAK-STAT signaling may impair interferon-mediated antiviral immune surveillance, potentially facilitating virus-related oncogenesis in immunosuppressed settings. Although direct evidence specifically linking JAK inhibitors to human papillomavirus- or Epstein–Barr virus-related cutaneous malignancies remains limited, the occurrence of viral-induced skin cancers is well recognized in the context of immunosuppression [18, 111, 112].

In contrast to conventional JAK inhibitors, TYK2 inhibitors such as deucravacitinib may provide a more selective mechanism of action, potentially translating into a more favorable oncologic safety profile. Clinical trials and long-term extension studies in dermatologic populations have reported low rates of malignancies, including non-melanoma skin cancer, supporting a reassuring cutaneous safety signal, although longer-term data are still needed to confirm these findings [42, 43, 113, 114].

5.1 Spectrum of Cutaneous Malignancies

Most cutaneous malignancies reported during JAK inhibitor therapy are NMSC (Fig. 8), particularly cutaneous squamous cell carcinoma (cSCC) and to a lesser extent basal cell carcinoma (BCC) [58, 115]. Available long-term datasets and extension studies suggest that the relative distribution of NMSC subtypes may differ from that observed in the general population, with a comparable or higher proportion of cSCC compared with BCC. In the ORAL Surveillance trial of RA, 37 cSCC and 35 BCC were reported in the combined tofacitinib group (SCC:BCC ratio 1.06:1), increasing to 22:16 (1.38:1) at the higher 10-mg twice-daily dose [116]. Similar patterns have been described in hematologic settings treated with ruxolitinib, where cSCC appears to predominate over BCC [109, 117].

Melanoma has been reported less consistently, limiting definitive conclusions regarding causality [58, 109]. In dermatologic clinical trials of oral JAK inhibitors, NMSC events are generally very rare, with reported frequencies that ranged from none to approximately 0.6–0.9% (Table 1) and often occurring in patients with pre-existing risk factors such as prior NMSC or extensive actinic damage [1, 56, 88]. These low absolute numbers suggest that, in dermatology populations with relatively short follow-up and younger age, cutaneous malignancies remain uncommon [1, 56, 61].

5.2 Drug- and Dose-Related Signals

The non-selective JAK inhibitor tofacitinib has been most extensively scrutinized for malignancy risk following the results of the ORAL Surveillance trial in RA, a large phase III–IV randomized study including 4362 patients aged ≥ 50 years with at least one additional cardiovascular risk factor. Patients were randomized to tofacitinib 5 or 10 mg twice daily or TNF inhibitors and followed for a median of approximately 4 years. The trial showed a higher incidence of malignancies excluding NMSC with tofacitinib compared with TNF inhibitors (hazard ratio 1.48, 95% confidence interval 1.04–2.09), as well as an increased risk of NMSC (hazard ratio 2.02, 95% confidence interval 1.17–3.50) [116]. Notably, the divergence in malignancy risk between tofacitinib and TNF inhibitors became apparent after approximately 18 months of treatment, and lung cancer emerged as the most frequent malignancy excluding NMSC in tofacitinib-treated patients, particularly at the 10-mg twice-daily dose [116]. However, systematic reviews and meta-analyses focusing specifically on tofacitinib across

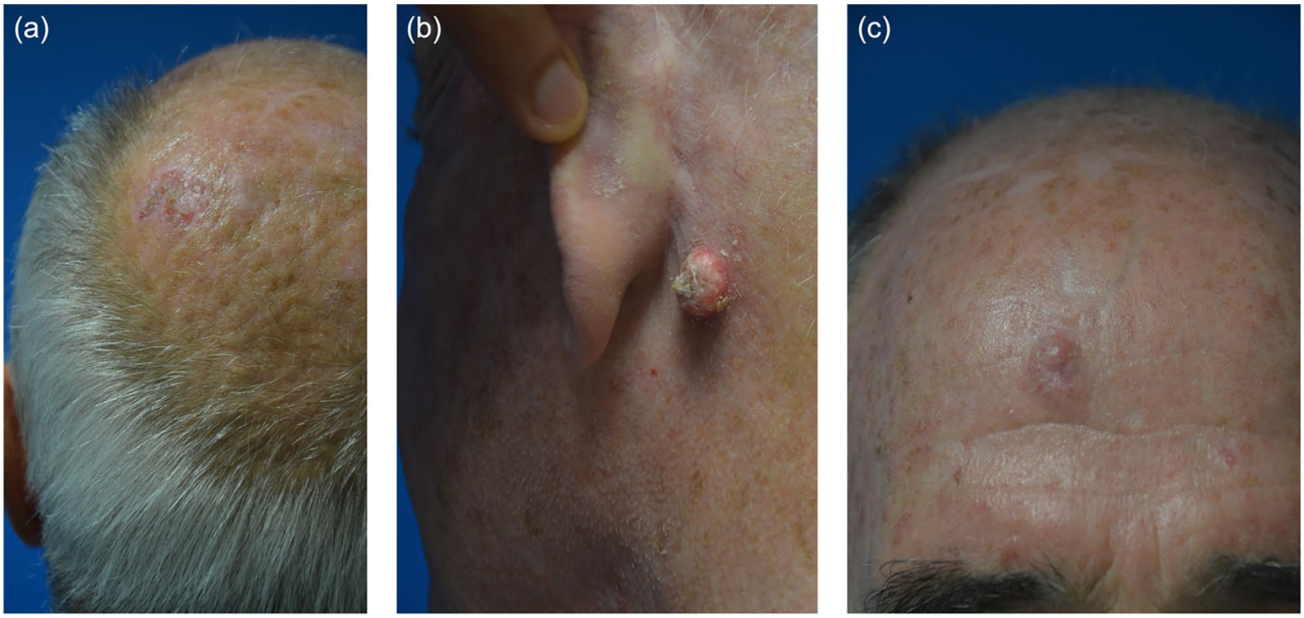

Fig. 8 Cutaneous squamous cell carcinoma of the scalp in a patient treated with ruxolitinib for 4 years (**a**). Retroauricular keratoacanthoma (**b**) and cutaneous squamous cell carcinoma of the forehead (**c**) in a patient receiving ruxolitinib. Personal files

randomized controlled trials have not consistently demonstrated a statistically significant increase in NMSC compared with control groups [10, 59, 118]. Isolated cases of NMSC have been reported, often in patients with previous skin cancer history, but no consistent increase in overall cancer incidence has been demonstrated to date [10, 59].

Janus kinase inhibitors seem to be associated with a lower oncologic signal in dermatologic settings, and long-term extension studies in pediatric AD have not reported malignancies over several years of follow-up, although the limited exposure time and young age of participants must be considered [119]. Similarly, pooled analyses and long-term extension studies of commonly used JAK inhibitors in dermatology, including abrocitinib, upadacitinib, and baricitinib, have not demonstrated a consistent increase in malignancy rates, including non-melanoma skin cancer, although the follow-up duration remains relatively limited and continued surveillance is warranted [13–15, 25, 31, 61, 88]. In addition, dose-dependent safety signals have been reported for commonly used JAK inhibitors in dermatology. For example exposure-adjusted event rates of malignant neoplasms were higher with upadacitinib 30 mg compared with 15 mg (0.9 vs 0.6 per 100 patient-years), supporting a potential dose-related increase in risk [120].

5.3 Ruxolitinib and High-Risk Hematologic Settings

A more consistent signal for cutaneous malignancies emerges in hematologic populations treated with first-generation JAK inhibitor ruxolitinib, particularly patients with polycythemia vera or myelofibrosis [57, 121–124]. An increased incidence of cSCC in these patients compared with unexposed controls, with some reports describing multiple or recurrent lesions is reported [121]. In addition to higher incidence, qualitative differences in tumor behavior have been described. Locally aggressive cSCC with poor histopathological differentiation and a high risk of recurrence were commonly reported during ruxolitinib therapy, raising the hypothesis that JAK1/2 inhibition may impair antitumor immune surveillance in a population already characterized by advanced age, cumulative ultraviolet exposure, and prior cytoreductive therapies (including hydroxyurea) [109, 125]. Similarly, in large series of patients with myeloproliferative neoplasms, ruxolitinib-associated NMSC represented the primary cause of death, exceeding deaths related to myelofibrosis [124].

In the same way, in the hematopoietic stem cell transplantation setting, where baseline risk of skin cancer is markedly elevated, exposure to ruxolitinib for the management of graft-versus-host disease has been associated with a high burden of NMSC compared with other immunosuppressive therapies, particularly cSCC and carcinoma in situ, although disentangling drug-specific effects from long-term transplant-related immunosuppression remains challenging [117, 126, 127]. The risk is inconclusive for basal cell carcinoma or melanoma [117].

Kaposi's sarcoma represents a rare but clinically relevant cutaneous malignancies reported during treatment with ruxolitinib [55]. Most reported cases involve elderly hematologic patients receiving ruxolitinib, in whom cutaneous or mucocutaneous violaceous papules and nodules developed after drug initiation (Fig. 9), often in the absence of known human immunodeficiency virus infection or other classical risk factors [55, 58, 128]. Similar lesions have been also described sporadically with tofacitinib or upadacitinib in patients treated for severe underlying disorders (e.g., RA, ulcerative colitis) [129, 130]. The pathogenesis is thought to be promoted by the loss of T-helper 1-driven immunity and the important role of JAK pathways in the proliferation of CD4+ T cells. The consequence is a secondary reactivation of human herpesvirus 8 following JAK pathway inhibition, particularly in patients with pre-existing immune dysfunction due to underlying disease [58].

5.4 Clinical Implications

The US Food and Drug Administration issued a black-box warning for all JAK inhibitors (with the exception of TYK2 inhibitor deucravacitinib) in patients presenting with high-risk factors for malignancy. However, available evidence suggests that the risk of cutaneous malignancies during JAK inhibitor therapy is not uniform across drugs or indications, but is strongly influenced by prior exposure to immunosuppressive therapies, age, patient-related factors, and disease context [60]. In dermatologic and IBD populations, where patients are often younger and less heavily immunosuppressed, the risk of developing skin cancers appear to be minimal and frequently associated with pre-existing risk factors (Table 4) [1, 59, 60, 86]. To date, no clear increase in cancer incidence with JAK inhibitors has been demonstrated in randomized clinical trials or real-world studies conducted in patients with AD or psoriasis [61]. In contrast, in hematologic settings, particularly with ruxolitinib, a more consistent association with NMSC, especially cSCC with an aggressive behaviour, has been observed [117, 121, 125].

Fig. 9 Kaposi sarcoma of the foot in a patient receiving ruxolitinib for myelofibrosis. Personal files

Overall, the immunosuppressive properties of JAK inhibitors and their negative effects on natural killer cells may promote carcinogenesis in specific situations depending on the presence of certain associated contributing factors [59, 106]. Therefore, it can be suggested that approved JAK inhibitors (except for deucravacitinib) should be prescribed with caution to patients with a current, previous, or increased risk of NMSC, and only when no other adequate therapeutic alternatives are available. Shared decision making should be clearly prioritized, taking into account the overall medical condition, the disease being treated, the benefit-risk ratio, and the associated risk factors [1, 124].

Finally, routine skin cancer surveillance with dermatologic examinations is advisable for all patients exposed, particularly in selected populations with prior NMSC, extensive actinic damage, older age, hematologic settings, or ruxolitinib exposure, and temporary interruption or discontinuation should be considered in the case of aggressive, recurrent, or multiple cSCC [58, 117, 125].

6 Topical JAK Inhibitors

So far, this review has mainly addressed dermatologic AEs associated with systemic JAK inhibitors. In contrast, topical JAK inhibitors are characterized by minimal systemic exposure and a distinct safety profile. Treatment is generally recommended for limited body surface area involvement, typically not exceeding 10–20% of body surface

Table 4 Main risk factors for cutaneous malignancies during JAK inhibitor therapy

Domain	Key risk factors
Patient related	Older age; fair phototype; cumulative UV exposure; prior history of non-melanoma skin cancer
Disease related	Hematologic disorders; chronic inflammatory diseases; baseline immune dysregulation
Treatment related	Long-term exposure; higher JAK inhibitor doses; concomitant or prior immunosuppressive therapies
Environmental factors	Chronic sun exposure; high UV index regions
Additional factors	Cumulative immunosuppression; impaired immune surveillance; possible contribution of oncogenic viruses

JAK Janus kinase, *NMSC* non-melanoma skin cancer, *UV* ultraviolet area, to minimize systemic absorption [17]. This is particularly relevant in chronic inflammatory dermatoses, where impaired skin barrier function may increase percutaneous drug absorption. The following section therefore focuses on the dermatologic safety of topical JAK inhibitors.

6.1 Topical Ruxolitinib

The safety data for topical JAK1/2 inhibitor ruxolitinib are primarily based on phase III data in AD and vitiligo, over a maximum period of 12 months [17, 131, 132]. The bioavailability of the product appears to be low, limiting a priori the risk of systemic toxicities reported with other JAK inhibitors used orally [131, 133, 134].

The main dermatological manifestations reported include application-site reactions, which remain limited (< 5–10%) and mild to moderate in intensity. It is possible to differentiate application-site acne (5.5%), pruritus (5%), rash/dermatitis (2%), and exfoliation (1%) [17, 131–133, 135]. The risk of localized superinfection, particularly viral (HSV, molluscum contagiosum), remains to be determined but the incidence appears to be low [132, 136]. In addition, no increased risk of herpes zoster has been reported [134].

Finally, there is no warning regarding the occurrence of secondary skin neoplasms [131, 133, 134, 137], including in higher-risk populations [136], with no clear connection to topical application of ruxolitinib. However, prospective monitoring should nevertheless be continued.

6.2 Topical Delgocitinib

Delgocitinib is the first-in-class, topical pan JAK inhibitor, acting by downregulation of the JAK dependent kinase signaling and modulation of T-helper 1 and T-helper 2 pathways. It is now European Medicines Agency and Food and Drug Administration approved for the management of refractory moderate-to-severe chronic hand eczema [138–140].

Delgocitinib presents a favorable dermatological safety profile. During the pivotal DELTA 1 and DELTA 2 clinical trials, application-site AEs were infrequent and reported by a comparable incidence of patients treated with delgocitinib cream and cream vehicle [138, 139]. Hand dermatitis, dry skin, and contact dermatitis were reported in about 1% of patients exposed, with a similar incidence in patients treated with vehicle or oral acitretin. Most skin reactions were mild to moderate in intensity. No allergic contact dermatitis was documented. Finally, no observation of HSV superinfection has been reported and application-site folliculitis/acne, impetigo, or eczema herpeticum has been uncommonly described [140–142].

7 Conclusions

Janus kinase inhibitors represent a major therapeutic advance across multiple dermatologic and non-dermatologic pathologies; however, their expanding use has highlighted a distinctive spectrum of dermatologic AEs. Among these, JAKne emerge as the most frequent, followed by infectious complications, particularly herpes zoster, and, less commonly, cutaneous malignancies.

Although long-term studies evaluating safety are scarce [56, 57, 88], available evidence suggests that most dermatologic AEs associated with JAK inhibitors are manageable, rarely lead to permanent treatment discontinuation, and often reflect an interaction between drug-related immunomodulation and patient-specific risk factors rather than a uniform class toxicity. Awareness and early recognition of these AEs are essential to optimize patient counseling, minimize impact on quality of life, and support long-term treatment adherence. Finally, long-term studies evaluating safety remain necessary.

Future research should also explore dermatologic AEs associated with JAK inhibitors in off-label dermatologic indications, beyond currently approved diseases. Expanding the evidence base across a broader range of inflammatory dermatoses may help identify disease-specific patterns of cutaneous toxicity and clarify the role of underlying pathogenic pathways in modulating AE profiles [1, 2, 44–49, 51]. In addition, novel approaches targeting the JAK-STAT pathway beyond oral and topical formulations are currently under development, including RNA interference-based therapies such as the JAK1-targeting siRNA ALY-101, currently evaluated in a phase IIa clinical trial for a trans-epidermal formulation for AA [143]. However, published clinical data remain limited, and the spectrum of dermatologic AEs associated with these agents is not yet well characterized, warranting further investigation.

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Declarations

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Ethics Approval Not applicable.

Consent to Participate Not applicable.

Consent for Publication The patients, or patient's parents/guardians, in this article have given written informed consent to the publication of their photos, according to the Declaration of Helsinki principles.

Availability of Data and Material No datasets were generated or analyzed during the current study. A review of the literature was conducted using previously published data.

Code Availability Not applicable.

Authors' Contributions All authors contributed to the conception and drafting of the manuscript and provided clinical images. All authors critically revised the manuscript for important intellectual content and approved the final version of the manuscript.

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